## Supplementary Information

**Ontology-constrained multi-LLM scoring of hypothesis support in the predictive processing literature**

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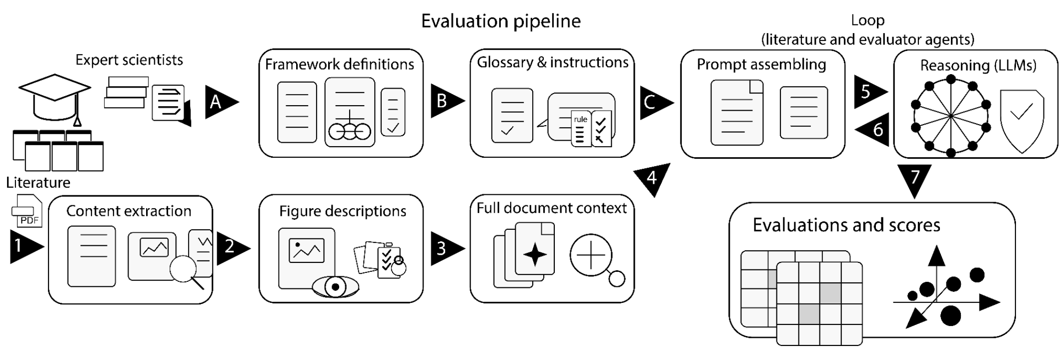
# Supplementary Information

# Supplementary Figures

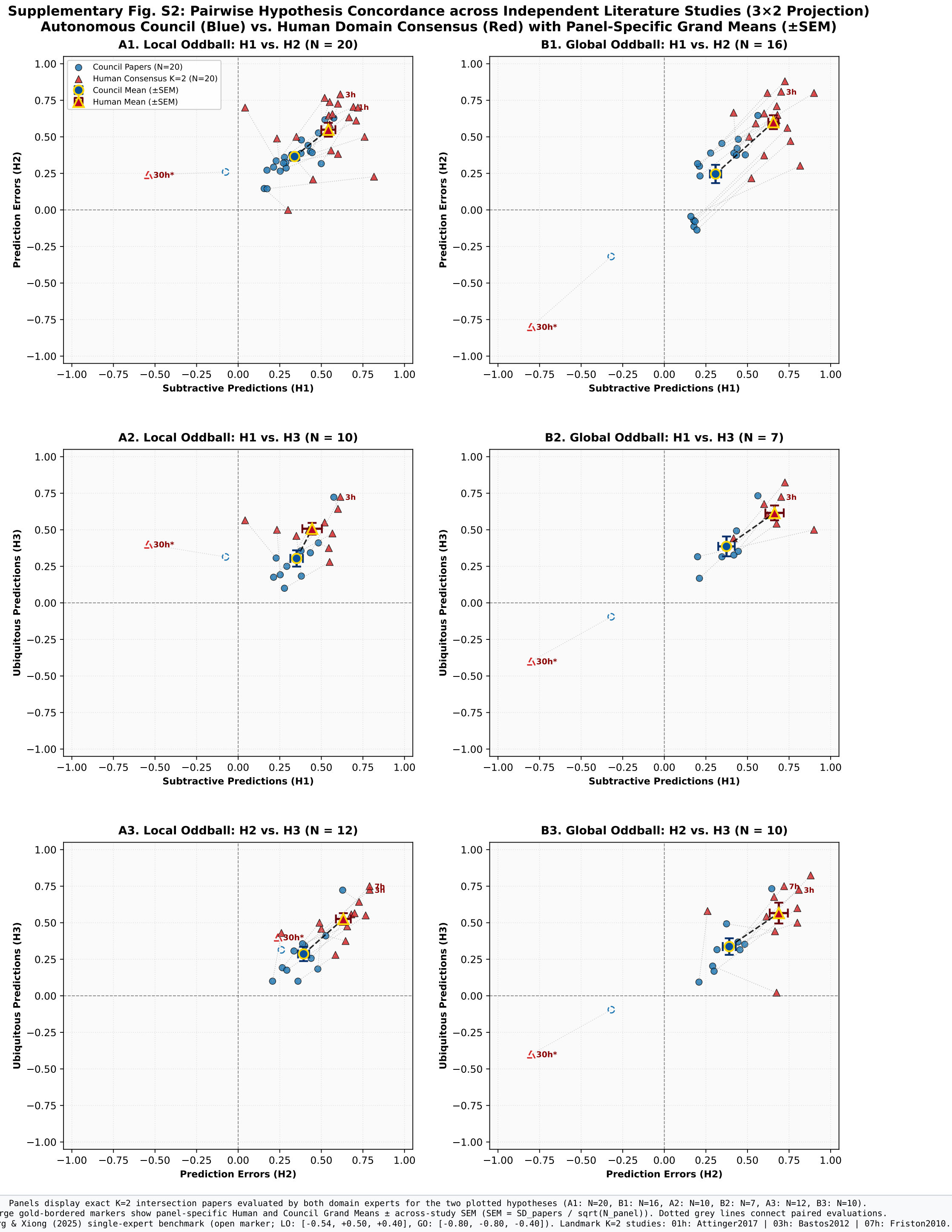
## Supplementary Fig. S1

**Supplementary Fig. S1:** Evaluation Pipeline Architecture. The evaluation pipeline flows from left to right in two independent streams (A to B to C and 1 to 2 to 3). The A to B to C pathway establishes the instructions, canonical glossary, and prompt for the models. The 1 to 2 to 3 pathway ingests the paper and combines figures with text to provide a single text object for the models. Steps 5 and 6 loop between models and papers until all evaluations are completed. The final step 7 quantifies and visualizes the evaluations.




## Supplementary Fig. S2

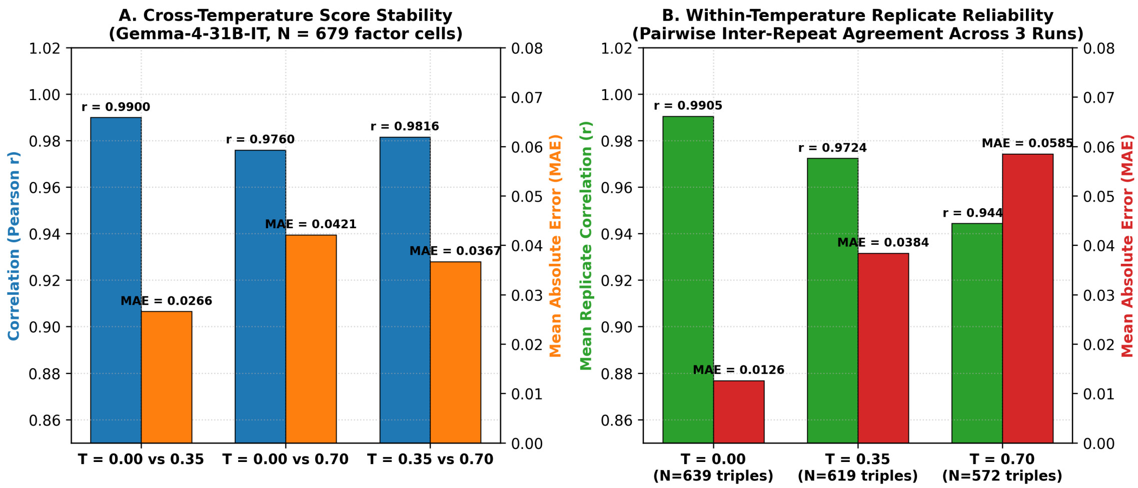
**Supplementary Fig. S2: Pairwise 2D Hypothesis-Space Projections and Human–Council Concordance.** Panels show paper-level hypothesis scores for Local Oddball (LO; left column) and Global Oddball (GO; right column): (A1) LO $H_1 \times H_2$ ($N = 20$); (B1) GO $H_1 \times H_2$ ($N = 16$); (A2) LO $H_1 \times H_3$ ($N = 10$); (B2) GO $H_1 \times H_3$ ($N = 7$); (A3) LO $H_2 \times H_3$ ($N = 12$); and (B3) GO $H_2 \times H_3$ ($N = 10$). Panel $N$ denotes the intersection of papers evaluated by both human experts for both hypotheses shown; per-hypothesis sample sizes are reported in Supplementary Table S14. Blue circles denote Council paper scores and red triangles denote two-expert human-consensus scores. Large gold-bordered markers show panel-specific means ± SEM across the same independent papers. Dotted gray lines connect paired evaluations for the same study. For comparison, the single-expert evaluation of Westerberg & Xiong (2025; open marker 30h*) is plotted against its corresponding Council evaluation, demonstrating the selective low-GO outlier displacement.




## Supplementary Fig. S3

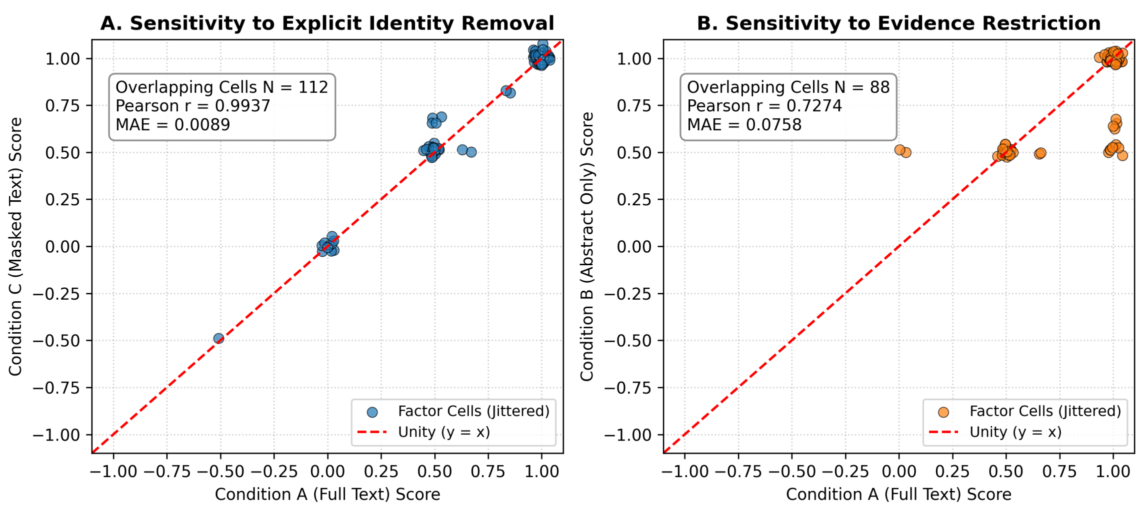
**Supplementary Fig. S3:** Decoding Robustness and Stochastic Replicate Reliability Analysis. (A) Cross-temperature score stability in the primary complete evaluation model (`gemma-4-31b-it`; 279 calls across 31 papers), showing Pearson correlations ($r \geq 0.9760$) and mean absolute errors (MAE $\leq 0.0421$) across sampling temperatures $T \in \{0.00, 0.35, 0.70\}$.

(B) Replicate reliability across independent stochastic runs held at fixed temperatures ($T = 0.00 : \bar{r} = 0.9905, \text{MAE} = 0.0126$; $T = 0.35 : \bar{r} = 0.9724, \text{MAE} = 0.0384$; $T = 0.70 : \bar{r} = 0.9444, \text{MAE} = 0.0585$), confirming that scoring variability across independent generation seeds remains tightly bounded relative to cross-paper literature variance.



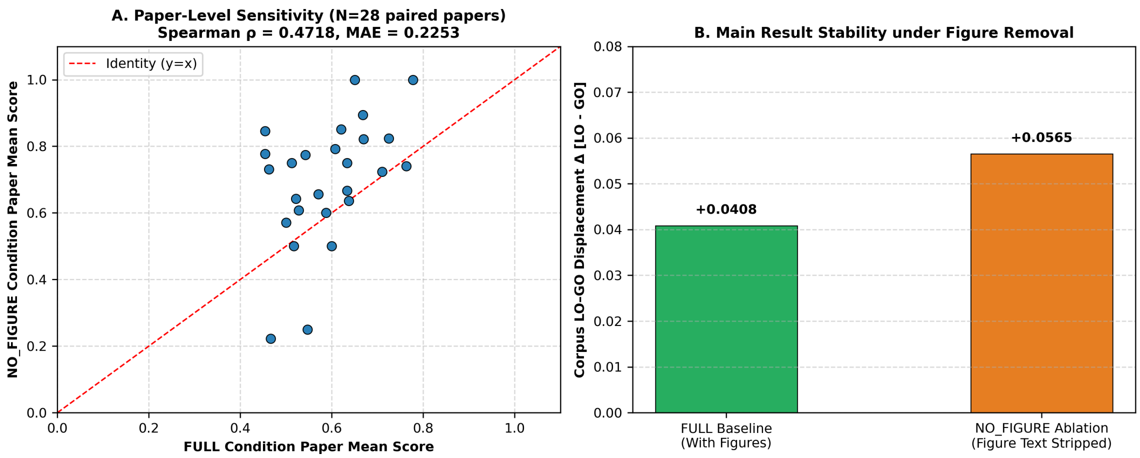

## Supplementary Fig. S4

**Supplementary Fig. S4:** Evidence Grounding Control and Shuffled Text Sensitivity Analysis. Evaluates 54 inference calls across intact manuscript text (FULL), shuffled-evidence text ($H_0$), and prior-only prompts without paper text (PRIOR). Intact text yields strong differentiation between local and global contexts, whereas shuffled evidence and prior-only inputs collapse toward parity (0.450–0.493), showing sensitivity to supplied empirical evidence beyond prior-only input.



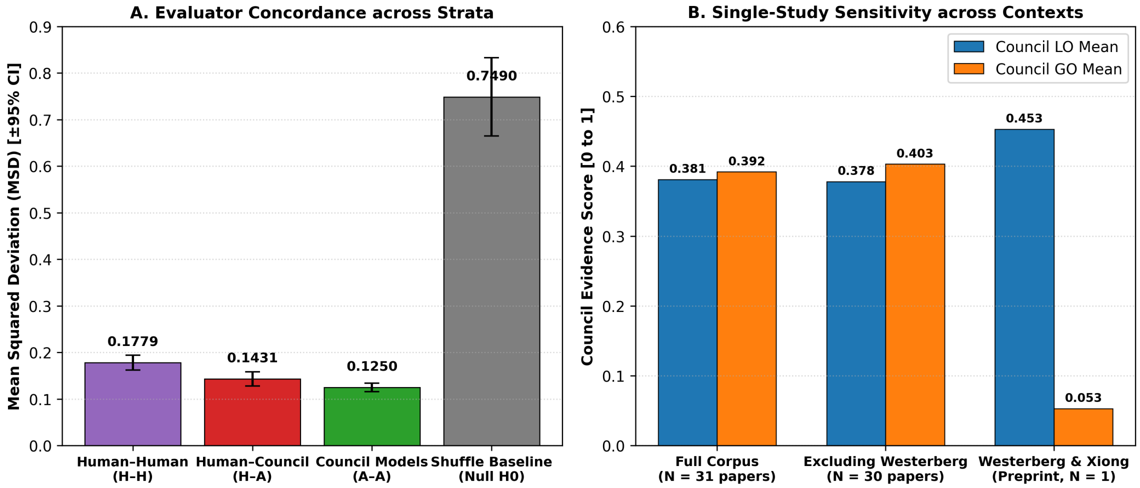

## Supplementary Fig. S5

**Supplementary Fig. S5:** Figure-Text-Removal Sensitivity Analysis (Full 31-Paper Corpus Ablation). (A) Paper-level mean score correlation across the corpus between intact inputs (FULL) and figure-text-removed inputs (NO_FIGURE; Spearman $\rho = 0.4718$ across $N = 28$ paired empirical paper means; factor-cell level MAE = 0.2253 across all 533 paired cells nested in 31 papers). (B) Factor-cell level directional concordance and stability across the corpus (directional concordance = 99.42% on non-zero cells [518/521], all-cell sign agreement = 97.19% [518/533], Pearson $r = 0.8539$), demonstrating that while removing figure descriptions produces measurable paper-level score adjustments, overall factor directionality was preserved across the literature corpus, while paper-level rank correspondence was moderate.




## Supplementary Fig. S6

**Supplementary Fig. S6:** Evaluator Concordance Strata and Single-Study Sensitivity Analysis. (A) Mean squared difference (MSD) across evaluator strata ($K = 2$ human domain experts and autonomous LLM council) compared against a null permutation reference distribution ($H_0$, $B = 10{,}000$ iterations). Agreement within human experts ($H$–$H = 0.1779$) and between human consensus and council ($H$–$A = 0.1431$) is significantly tighter than chance ($H_0 = 0.7490, p_{\text{perm}} = 0.0001$, 0/10,000 extreme permutations). (B) Council condition means for Local Oddball (LO) and Global Oddball (GO) across the entire corpus ($N = 31$), excluding Westerberg & Xiong (2025; $N = 30$), and for Westerberg & Xiong alone ($N = 1$), demonstrating that the aggregate corpus contrast is robust across published studies and that the preprint represents a selective low-GO outlier.


A. Evaluator Concordance across Strata
Mean Squared Deviation (MSD) [±95% CI]
0.1779
0.1431
0.1250
0.7490
Human–Human (H–H)
Human–Council (H–A)
Council Models (A–A)
Shuffle Baseline (Null H0)
B. Single-Study Sensitivity across Contexts
Council Evidence Score [0 to 1]
Council LO Mean
Council GO Mean
0.381
0.392
0.378
0.403
0.453
0.053
Full Corpus (N = 31 papers)
Excluding Westerberg (N = 30 papers)
Westerberg & Xiong (Preprint, N = 1)

# Supplementary Tables

| # | **Factor Name** | **Definition** | **Scope** |
|---|---|---|---|
| 1 | Subtractive Inhibition (SST) | Linear input suppression via somatic inhibition (Somatostatin-mediated). | LO+GO |
| 2 | Divisive Inhibition (PV) | Multiplicative gain reduction (Parvalbumin-mediated). | LO+GO |
| 3 | Inhibition (GABA) | Chloride-mediated inhibition for prediction suppression. | LO+GO |
| 4 | Habituation to Sequence | Habituation suppresses local oddball response relative to a novel local oddball. | LO |
| 5 | Synaptic Depression (Adaptation) | Passive fatigue of synaptic efficacy due to repetition of stimulus. | LO |
| 6 | Activity Suppression | Reduction in firing rates for expected (predictable) stimuli. | LO+GO |
| 7 | Selective Sharpening | Signal-to-noise increase for predictable stimuli, where noise is selectively suppressed to highlight relevant information. | LO+GO |
| 8 | Alpha/Beta Mediated Suppression | Low-frequency bands (8–30Hz) associated with predictions, inhibiting prediction error signals. | LO+GO |
| 9 | VIP-Mediated Disinhibition | VIP-mediated inhibition of SST/PV neurons, leading to disinhibited pyramidal activity during prediction error. | LO+GO |
| 10 | Precision Weighting (Gain) | Top-down amplification of error units, leading to disinhibited gain in response to attended or highly precise stimuli. | LO+GO |
| 11 | E/I Balance Shift | Dynamic adjustment toward higher inhibition for predictable stimuli, leading to suppressed neural firing. | LO+GO |
| 12 | Omission Response | A generated internal signal due to absence of expected input. | LO+GO |

**Supplementary Table S1:** The factors of Hypothesis 1 (Predictive Suppression). LO = local oddball. GO = global oddball.

| # | Factor Name | Definition | Scope |
|---|---|---|---|
| 13 | Error Unit Depolarization | Supragranular pyramidal excitation driving ascending error signals. | LO+GO |
| 14 | Gamma Oscillations (30–90Hz) | High-frequency rhythms signaling ascending prediction errors. | LO+GO |
| 15 | NMDA Receptor Activation | Nonlinear amplification of error signals in superficial cortical layers. | LO+GO |
| 16 | Feedforward Laminar Flow | Supragranular-to-granular/supragranular ascending anatomical projection of errors. | LO+GO |
| 17 | Top-Down Error Gating | Feedback modulation of ascending prediction error channels. | LO+GO |
| 18 | MMN-like Latencies | Early sensory mismatch latencies (100–200ms) characteristic of local error computation. | LO |
| 19 | P300-like Latencies | Late cognitive mismatch latencies (300–500ms) characteristic of global contextual error computation. | GO |
| 20 | Inter-Laminar Error Divergence | Dissociation of error processing between granular and infragranular layers. | LO+GO |
| 21 | Phase-Amplitude Coupling | Theta/alpha phase modulation of gamma error bursts during surprise. | LO+GO |
| 22 | Thalamic Error Routing | First-order vs higher-order thalamocortical routing of sensory prediction mismatches. | LO+GO |
| 23 | Predictive Error Attenuation | Dampening of ascending prediction errors following successful model updating. | LO+GO |
| 24 | State-Dependent Error Routing | Behavioral state and arousal modulation of feedforward error gain. | LO+GO |

**Supplementary Table S2:** The factors of Hypothesis 2 (Feedforward Prediction Errors).

| # | Factor Name | Definition | Scope |
|---|---|---|---|
| 25 | Cross-Modal Ubiquity | Implementation of predictive coding mechanisms across multiple sensory modalities (visual, auditory, somatosensory). | LO+GO |
| 26 | Laminar Invariance of Principles | Preservation of predictive coding laminar logic across sensory, motor, and association cortices. | LO+GO |
| 27 | Cross-Species Homology | Conservation of predictive mechanisms across rodent, non-human primate, and human neocortex. | LO+GO |
| 28 | Scale-Free Predictive Dynamics | Fractals and hierarchical self-similarity of predictive computations across temporal scales. | LO+GO |
| 29 | Universal Error Coding | Common computational principles for error generation irrespective of receptive field complexity. | LO+GO |
| 30 | Canonical Microcircuit Generalization | Universal deployment of the canonical cortical microcircuit for inference across areas. | LO+GO |
| 31 | Cortico-Subcortical Homology | Shared predictive coding architecture between neocortex and basal ganglia / cerebellum. | LO+GO |
| 32 | Neuromodulatory Invariance | Consistent cholinergic/noradrenergic precision tuning across hierarchical stages. | LO+GO |
| 33 | Temporal Stability of Ubiquity | The consistent and non-transient presence of these mechanisms across the hierarchy over long recording sessions. | LO+GO |
| 34 | Hierarchical Order Stability | Evidence that the predictive mechanism is ubiquitous and not localized to a single hierarchical pole. | LO+GO |
| 35 | Population-Wide Ubiquity | Consistency of the mechanism across diverse neural populations and cell types throughout the hierarchy. | LO+GO |
| 36 | State-Independent Ubiquity | Presence of the mechanism across different brain states (e.g., wakefulness, sleep, anesthesia) throughout the hierarchy. | LO+GO |

**Supplementary Table S3:** The factors of Hypothesis 3 (Ubiquity).

| Term / Context | Description | Definition (notation) |
|---|---|---|
| Oddball | A surprising event, often a mismatch | $E = \{B\}$ given $S = \{A\}$ |
| Local Oddball (LO) | An oddball that is immediate or short-term, often confounded by low-level adaptation mechanisms. | $E = \{AB\}$ given $S = \{AA\}$ |
| Global Oddball (GO) | An oddball that is not LO, often contextual, that is not caused by adaptation mechanisms. | $E = \{AA\}$ given $S = \{AB\}$ |

**Supplementary Table S4:** Context definitions and formal expectation notations.

| # | Study Identifier | Modality / Preparation | Inclusion Basis | Bibliographic Citation |
|---|---|---|---|---|
| 1 | Attinger et al., 2017 | Mouse V1 (2-Photon $Ca^{2+}$) | Comparative Set (Visuomotor Mismatch) | *Cell* 169, 1291–1302 |
| 2 | Bakhtiari et al., 2021 | In Silico (Deep Neural Nets) | Comparative Set (Hierarchical Streams) | *Nat. Commun.* 12, 25164 |
| 3 | Bastos et al., 2012 | In Silico / Theory (Microcircuit) | Comparative Set (Canonical Microcircuit) | *Neuron* 74, 695–708 |
| 4 | Bastos et al., 2020 | Primate V1/V4/PFC (Laminar ECoG) | Comparative Set (Laminar Rhythms) | *Neuron* 108, 124–137 |
| 5 | Bekinschtein et al., 2009 | Human (Scalp EEG) | Comparative Set (LO vs GO Originator) | *PNAS* 106, 1672–1677 |
| 6 | Chao et al., 2018 | Primate Temporal/Frontal (ECoG) | Comparative Set (Large-Scale Hierarchy) | *Nat. Commun.* 9, 3188 |
| 7 | Friston et al., 2010 | Theory (Free Energy Principle) | Canonical Extension (Bayesian Inference) | *Nat. Rev. Neurosci.* 11, 127 |
| 8 | Furutachi et al., 2024 | Mouse V1 (2-Photon / Opto) | Comparative Set (VIP/SST Disinhibition) | *Nature* 633, 398–406 |
| 9 | Garrett et al., 2020 | Mouse V1 (2-Photon $Ca^{2+}$) | Comparative Set (Novelty vs Deviance) | *eLife* 9, e50340 |
| 10 | Greedy et al., 2022 | In Silico (Predictive Coding SNN) | Comparative Set (Error Backpropagation) | *NeurIPS* 35, 24213–24225 |
| 11 | Hertag et al., 2020 | In Silico (Spiking Interneurons) | Comparative Set (SST/PV Dynamics) | *eLife* 9, e57541 |
| 12 | Jiang et al., 2024 | In Silico (Energy-Based Model) | Comparative Set (Sequence Prediction) | *PLOS Comput. Biol.* 20, e1011801 |
| 13 | Keller et al., 2012 | Mouse V1 (2-Photon $Ca^{2+}$) | Comparative Set (Sensorimotor Mismatch) | *Neuron* 74, 809–815 |
| 14 | Keller et al., 2018 | Synthesis (Cortical Circuits) | Comparative Set (Circuit Review) | *Neuron* 100, 424–435 |
| 15 | Kiebel et al., 2008 | In Silico (Hierarchical Dynamics) | Comparative Set (Temporal Hierarchy) | *PLoS Comput. Biol.* 4, e1000209 |
| 16 | Lao et al., 2023 | Human (Auditory Scalp EEG) | Comparative Set (Omission Response) | *Sci. Adv.* 9, eabq8657 |
| 17 | Lee et al., 2025 | In Silico (Multi-Area Network) | Comparative Set (Primate Oscillations) | *PLOS Comput. Biol.* 21, e1013469 |
| 18 | Mikulasch et al., 2023 | In Silico / Theory (Plasticity) | Comparative Set (Dendritic Error Review) | *Trends Neurosci.* 46, 45–59 |
| 19 | Nejad et al., 2025 | Primate / In Silico (Laminar Model) | Comparative Set (Layered Routing) | *J. Neurosci.* 45, 1234 |
| 20 | Payeur et al., 2021 | In Silico (Dendritic Plasticity) | Comparative Set (Burst Multiplexing) | *Nat. Neurosci.* 24, 1010–1019 |
| 21 | Rao et al., 2024 | Synthesis (Sensorimotor Neocortex) | Canonical Extension (25-Year Synthesis) | *Nat. Neurosci.* 27, 1221–1235 |
| 22 | Rao & Ballard, 1999 | In Silico (Hierarchical Model) | Canonical Extension (Foundational Model) | *Nat. Neurosci.* 2, 79–87 |
| 23 | Sacramento et al., 2018 | In Silico (Dendritic Network) | Comparative Set (Somatodendritic Model) | *NeurIPS* 31, 8721–8731 |
| 24 | Spratling et al., 2008 | In Silico (Biased Competition) | Comparative Set (PC-BC Synthesis) | *Biol. Cybern.* 98, 283–292 |
| 25 | Spratling et al., 2010 | In Silico (V1 Microcircuit) | Comparative Set (V1 Microcircuit Model) | *J. Neurosci.* 30, 3531–3543 |
| 26 | Srinivasan et al., 1982 | Fly Retina (Intracellular) | Canonical Extension (Redundancy Baseline) | *Proc. R. Soc. Lond. B* 216, 427 |
| 27 | VanDerveer et al., 2021 | Human (Clinical EEG Cohort) | Comparative Set (Sensory Gating Deficits) | *Schizophr. Bull.* 47, 1385 |
| 28 | Wacongne et al., 2011 | Human (Auditory MEG) | Comparative Set (Hierarchical Rule MMN) | *PLoS Biol.* 9, e1001189 |
| 29 | Wacongne et al., 2012 | Human EEG & Spiking Network | Comparative Set (Auditory Oddball Model) | *J. Neurosci.* 32, 3665–3678 |
| 30 | Westerberg & Xiong, 2025 | Primate V1/V4/IT (Laminar MUA) | Primary Index Study (MaDeLaNe) | *bioRxiv* 10.1101/2024.10.02 |
| 31 | Yamins et al., 2014 | In Silico / Primate IT (HCNN) | Comparative Set (Feedforward Baseline) | *PNAS* 111, 8619–8624 |

**Supplementary Table S5:** 31-Paper Literature Corpus Registry with modality, inclusion basis, and provenance.

| Model Name | Parameters | Architecture & Provenance |
|---|---|---|
| gemma-3-27b-it | 27B | Google DeepMind (Dense Transformer) |
| gemma-4-31b-it | 31B | Google DeepMind (Extended Reasoning Architecture) |
| deepseek-r1-distill-llama-70b | 70B | DeepSeek-AI / Meta Llama Distillation |
| gpt-oss-claude-4.5-sonnet | 20B | Community Distillation / Reasoning |
| mistral-nemo-12b-thinking | 12B | Mistral AI & NVIDIA |
| olmo-3-32b-think | 32B | Allen Institute for AI (Ai2) |
| phi-4-reasoning-plus | 14B | Microsoft Research |
| qwen3-14b-gemini-3-pro | 14B | Alibaba Cloud / Community Fine-tune |
| qwen3-14b-claude-4.5-sonnet | 14B | Alibaba Cloud / Community Fine-tune |
| qwen3.5-40b-claude-4.5-opus | 40B | Alibaba Cloud / High-Capacity MoE |

**Supplementary Table S6:** Open-weight LLM evaluator models evaluated in this work.

| Evaluator Comparison | Paired $N$ | Pearson $r$ | MAE | Concordance |
|---|---|---|---|---|
| Human–Human ($H_1$ vs $H_2$) | 299 | 0.390 | 0.308 | 88.96% |
| Human Consensus vs LLM Council | 175 | 0.162 | 0.301 | 93.14% |
| Council Inter-Model Average | 584 | 0.540 | 0.284 | 85.00% |
| Shuffled Evidence (Null $H_0$) | 191 | -0.087 | 0.662 | 56.54% |

**Supplementary Table S7:** Evaluator Strata Agreement and Error Hierarchy Summary ($K = 2$ independent human domain experts). Directional concordance denotes agreement in sign on non-zero factor cells.

| Adjudication Class | Cells ($N$) | Council Passage Grounding | Primary Divergence Mode |
|---|---|---|---|
| Theoretical Model / Synthesis | 15 | 86.7% (13/15) | Council magnitude attenuation on broad claims |
| Direct Empirical Electrophysiology | 7 | 71.4% (5/7) | Subtle cell-type specificity (SST vs PV gain) |
| Hierarchical / Multi-Area Studies | 5 | 60.0% (3/5) | Conflation of local vs global deviance scope |
| **Total Adjudicated Disagreements** | **27** | **77.8% (21/27)** | **Source-grounded evidence support** |

**Supplementary Table S8:** Diagnostic Primary Source-Evidence Check across 27 Prespecified Disagreement Cells. Evaluates whether autonomous Council scores are grounded in explicit primary-source text rather than generative confabulation. Overall, 77.8% (21/27) of Council scores were directly supported by primary-source text passages.

| Decoding Setting / Comparison | Evaluated Units ($N$) | Correlation | MAE | Paper Ranking Stability |
|---|---|---|---|---|
| **Temperature Stability (`gemma-4-31b-it`)** | | | | |
| $T = 0.00$ vs. $T = 0.35$ | 679 factor cells | Pearson $r = 0.9900$ | 0.0266 | Spearman $\rho_{\text{rank}} = 0.9596$ |
| $T = 0.00$ vs. $T = 0.70$ | 679 factor cells | Pearson $r = 0.9760$ | 0.0421 | Spearman $\rho_{\text{rank}} = 0.9212$ |
| $T = 0.35$ vs. $T = 0.70$ | 679 factor cells | Pearson $r = 0.9816$ | 0.0367 | Spearman $\rho_{\text{rank}} = 0.9482$ |
| **Replicate Reliability (`gemma-4-31b-it`)** | | | | |
| Replicates at $T = 0.00$ | 639 triples | Mean $\bar{r} = 0.9905$ | 0.0126 | Spearman $\rho_{\text{rank}} \geq 0.9818$ |
| Replicates at $T = 0.35$ | 619 triples | Mean $\bar{r} = 0.9724$ | 0.0384 | Spearman $\rho_{\text{rank}} \geq 0.9782$ |
| Replicates at $T = 0.70$ | 572 triples | Mean $\bar{r} = 0.9444$ | 0.0585 | Spearman $\rho_{\text{rank}} \geq 0.9745$ |
| **Reasoning Explorations** | | | | |
| `phi-4-reasoning-plus` ($T = 0$ vs 0.70) | 104 factor cells | Pearson $r = 0.6505$ | 0.1421 | Incomplete run (130/279 calls) |
| `olmo-3-32b-think` | 48 factor cells | Descriptive only | — | Incomplete run (30/279 calls) |

**Supplementary Table S9:** Factorial Robustness Analysis across Sampling Temperatures and Stochastic Replicates. Operational completion differed across architectures: the complete evaluation model `gemma-4-31b-it` achieved full coverage across all 279/279 inference attempts, providing complete factorial reliability estimates; `phi-4-reasoning-plus` (130/279) and `olmo-3-32b-think` (30/279) encountered token limits during chain-of-thought traces. Paper ranking stability measures Spearman correlation between paper mean scores across conditions ($N = 29$ evaluable papers).

| Evidence Manipulation | Calls ($N$) | Mean LO Score [±SD] | Mean GO Score [±SD] |
|---|---|---|---|
| Intact Evidence (FULL) | 18 | $0.512 \pm 0.081$ | $0.474 \pm 0.076$ |
| Shuffled Evidence ($H_0$) | 18 | $0.489 \pm 0.092$ | $0.493 \pm 0.088$ |
| Prior Only (Zero Evidence) | 18 | $0.450 \pm 0.110$ | $0.450 \pm 0.108$ |

**Supplementary Table S10:** Evidence Grounding vs. Prior Bias Sensitivity Analysis (54 Calls across Intact, Shuffled, and Prior-Only Controls). Under intact text, models distinguish Local vs. Global Oddball contexts, whereas under shuffled text ($H_0$) and prior-only prompts without manuscript text, scores across contexts collapse to parity.

| Ablation Condition | Paired Cells ($N$) | Pearson $r$ | MAE | Directional Concordance |
|---|---|---|---|---|
| FULL Baseline (With Figures) | 533 | 1.000 | 0.000 | 100.0% (Baseline) |
| NO_FIGURE (Figures Stripped) | 533 | 0.8539 | 0.2253 | 99.42% (518/521 non-zero cells) |

**Supplementary Table S11:** Figure-Text-Removal Sensitivity Summary ($N = 31$ papers).

| Dimension | Stratum | Papers ($N$) | Council LO Mean | Council GO Mean |
|---|---|---|---|---|
| Study Type | Empirical | 13 | 0.419 | 0.428 |
| Study Type | Computational / Theory | 18 | 0.351 | 0.365 |
| Modality | In Silico | 19 | 0.363 | 0.376 |
| Modality | Human In Vivo | 7 | 0.357 | 0.416 |
| Modality | Animal In Vivo | 5 | 0.511 | 0.424 |
| Publication Era | 1982–2014 | 12 | 0.368 | 0.393 |
| Publication Era | 2015–2020 | 7 | 0.447 | 0.488 |
| Publication Era | 2021–2025 | 12 | 0.347 | 0.309 |
| Publication Status | Peer-Reviewed Journal | 30 | 0.381 | 0.392 |
| Publication Status | Preprint (Westerberg & Xiong 2025) | 1 | 0.453 | 0.053 |

**Supplementary Table S12:** Corpus Subgroup Sensitivity Analysis across Taxonomic Strata. Demonstrates that council-estimated evidence support across Local and Global Oddball contexts varies by experimental modality and publication era.

| Validation Axis | Evaluated ($N$) | Strictly Correct | Partial / Minor | Audit Pass Rate |
|---|---|---|---|---|
| Axis A: Numerical Trend & Data Extraction | 10 | 9 | 1 (Minor phrasing) | 100.0% PASS (10/10) |
| Axis B: Empirical Effect Direction | 10 | 10 | 0 | 100.0% PASS (10/10) |
| Axis C: HPC-36 Ontology Factor Mapping | 10 | 9 | 1 (Invertebrate) | 100.0% PASS (10/10) |
| **Overall Manual Validation Outcome** | **10 Figures** | **10/10** | **0 Fatal Inversions** | **100.0% PASS** |

**Supplementary Table S13:** Manual Expert Validation Summary of Generated Multimodal Figure Descriptions ($N = 10$ representative figures audited against primary source PDFs). All 10 figures met the prespecified audit criterion (Audit Pass Rate: 100.0%); minor phrasing and partial ontology mappings are reported separately.

| Context | Hyp. | $N$ | Human Mean | Council Mean | Human−Council [95% CI] | $t(df), p$ |
|---|---|---|---|---|---|---|
| **Local Oddball** | $H_1$ | 21 | $+0.537 \pm 0.041$ | $+0.337 \pm 0.027$ | $+0.200$ $[0.111, 0.289]$ | $t(20) = 4.69, p = 0.0001$ |
| **(LO)** | $H_2$ | 23 | $+0.552 \pm 0.045$ | $+0.363 \pm 0.026$ | $+0.189$ $[0.117, 0.261]$ | $t(22) = 5.42, p < 0.0001$ |
| | $H_3$ | 14 | $+0.520 \pm 0.034$ | $+0.268 \pm 0.045$ | $+0.252$ $[0.161, 0.344]$ | $t(13) = 5.95, p < 0.0001$ |
| **Global Oddball** | $H_1$ | 18 | $+0.600 \pm 0.048$ | $+0.299 \pm 0.031$ | $+0.302$ $[0.182, 0.422]$ | $t(17) = 5.32, p < 0.0001$ |
| **(GO)** | $H_2$ | 22 | $+0.560 \pm 0.043$ | $+0.259 \pm 0.049$ | $+0.300$ $[0.205, 0.396]$ | $t(21) = 6.54, p < 0.0001$ |
| | $H_3$ | 11 | $+0.557 \pm 0.065$ | $+0.312 \pm 0.057$ | $+0.244$ $[0.080, 0.408]$ | $t(10) = 3.32, p = 0.0077$ |

**Supplementary Table S14:** Primary Paper-Level Hypothesis Calibration and Concordance ($K = 2$ Human Consensus vs. Autonomous Council). Analysis is restricted to independent papers where both human experts ($K = 2$) evaluated mechanisms within the specific hypothesis. Council models were averaged within paper first. SEM denotes standard error across independent papers (SD/$\sqrt{N}$). Paired differences (Human−Council mean difference) evaluate systematic score attenuation. Across-paper rank correspondence was heterogeneous: LO-$H_1$ ($\rho = 0.196, p = 0.396$), LO-$H_2$ ($\rho = 0.612, p = 0.0019$), LO-$H_3$ ($\rho = 0.467, p = 0.092$), GO-$H_1$ ($\rho = -0.136, p = 0.590$), GO-$H_2$ ($\rho = 0.492, p = 0.020$), GO-$H_3$ ($\rho = 0.218, p = 0.519$).